



DIGITAL TRANSFORMATION

UNDERSTANDING BUSINESS GOALS, RISKS, PROCESSES, AND DECISIONS

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Mathias Cöster, Mats Danielson, Love Ekenberg, Cecilia Gullberg, Gard Titlestad, Alf Westelius, and Gunnar Wettergren, *Digital Transformation: Understanding Business Goals, Risks, Processes, and Decisions*. Cambridge, UK: Open Book Publishers, 2023, <https://doi.org/10.11647/OBP.0350>

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ISBN Paperback: 978-1-80511-060-6

ISBN Hardback: 978-1-80511-061-3

ISBN Digital (PDF): 978-1-80511-062-0

ISBN Digital ebook (EPUB): 978-1-80511-063-7

ISBN XML: 978-1-80511-065-1

ISBN HTML: 978-1-80511-066-8

DOI: 10.11647/OBP.0350

Cover photo by Love Ekenberg

Cover design by Jeevanjot Kaur Nagpal

7. Probability and Risk Management

In Chapters 5 and 6, we went through various decision situations and considered how they might be modelled. However, uncertainty about the future affects most business decisions, since most of the world is simply uncertain. Although it is impossible to know anything about future scenarios with absolute certainty, uncertainty must somehow be modelled and managed because uncertain future events can greatly affect the outcome of a decision, and we will now investigate how to handle this complicating factor.

7.1. Probabilities and Decisions

Let us assume as discussed in the previous chapters that the main goals are defined and clearly communicated within the organisation. Let us also assume that the decision components, including the modelled scenarios, the possible consequences in each separate scenario, and the set of criteria, have been clarified during the decision process. Furthermore, we have a sense of how to prioritise the criteria and the values of the consequences, and we believe that we might know something about the probabilities of the different scenarios' outcomes. What should we do with all of this information? If we have structured the decision situation with all of the relevant components, how do we actually make a decision based on it?

An important aspect that is too rarely considered is how one determines what a correct decision is. So we need some kind of framework based on reasonable principles. One way to begin is to look

at how much we know about different scenarios and the probabilities that they will occur.

We usually distinguish between three main cases of uncertain decisions, even if most realistic situations reflect different mixtures of these categories:

- Strict uncertainty, where one does not know anything at all about the probabilities
- Uncertainty, where one does not know what will happen but does know the exact probability of each possible scenario
- Certainty, where one knows for sure which scenario will occur

Now assume that our organisation is in the middle of a digital transformation, and that we are considering whether or not we should streamline parts of the system development by introducing a new server platform. We have two strategies: i) to change platforms and ii) to maintain the current platform.

We are uncertain about the effects of a platform change. It is likely that it will take significant time for the staff to adapt to a new platform, which could mean that they experience the change negatively, at least initially. This, in turn, could lead to major reductions in efficiency. However, the members of staff are divided on this issue since a large proportion of them do not want to maintain the current platform. We therefore make the simplified assumption that there are two main scenarios of principal interest (in which we assume that other factors are the same): a) The staff react negatively and b) The staff react positively.

Depending on what we decide, and which scenario then unfolds, different consequences will follow. If we select the strategy of introducing a new platform and the staff reacts negatively, we will experience large efficiency losses. If the staff reacts positively to the platform change, we will have a clear increase in efficiency. Conversely, if we choose to keep the present platform and the staff reacts negatively, we will experience (comparatively small) efficiency losses, and if the staff reacts positively the business will continue as usual but less well than we would like. We can illustrate this via a so-called decision matrix, as in Table 7.1.

Table 7.1. A decision matrix.

	Negative staff	Positive staff
Change platforms	Large losses	Large gains
Keep the present	Small losses	Very small losses

Based on this matrix, we can now characterise the decision situation.

- In decisions made with certainty, there is only one state in the matrix. For example, we know for sure that whatever we do, the staff will react negatively.
- With strict uncertainty, we usually do not assign any probabilities at all, because we have no way of predicting them.
- In decisions with risk, we estimate probabilities of different outcomes. For example, we might estimate based on existing information, that the probability of the staff reacting negatively is 35% and that the probability that they will react positively is 65%.

These types of decision situations are solved differently. In decisions under security, the issue is about finding reasonable value estimates and then choosing the strategy with the best consequence. For example, suppose that we, for some reason, know for certain that staff will react negatively to whatever we do. In that case, the best thing is to keep the platform as it is since the losses will be smaller. Similarly, if we know for certain that staff will react positively, it is best to change platforms. In both cases, we only make the choice based on how beneficial the consequences are (and we assume that increased efficiency yields higher consequence values). The choice is thus pretty simple, and we just have to make up our minds.

When we have decisions under strict uncertainty, the situation is of course different. In these cases, we do not know anything about the future at all, so it is rather a matter of our relationship to risk. We sometimes consider the worst values of the consequences and choose the strategy that would yield the best consequence if the worst were to happen. In that case, we would therefore choose not to change

the platform because the worst consequence would be a large loss of efficiency, and we would prefer to exclude that strategy. This decision principle is called *maximin* and it is a highly defensive rule that advocates choosing the strategy whose worst consequence is the least bad one of all. We can simply search for the matrix row whose worst consequence is as good as possible. For the more optimistically inclined decision-maker, there is the *maximax* rule, which is instead aggressive: it dictates that we should choose the strategy whose best consequence is the best of all the consequences. There are also mixed forms of these decision rules and several additional suggestions depending on how much we would regret the decision if it turned out that we had chosen the wrong strategy. But it is not really worth dwelling on such details here since they are often only of theoretical interest. In practice, it is not really likely that in a real-life decision situation there would be neither information nor opinions regarding the probabilities available. For our purposes, we can therefore now leave decision rules under strict uncertainty.

Next, we have decisions under risk where we know something about the probabilities of the different consequences. This is often the case in, for example, market decisions and investments (and actually in most decision situations). Decisions under risk can also be modelled with a matrix model, where we then add a new row with probabilities for each state, which is thus assigned a probability (p_i), which is estimated in one way or another. Table 7.2 shows the (precise) probabilities of the scenarios and the (precise) values (v_i) of the consequences. We assume here that the values are on a scale from 0 (the worst) to 1 (the best).

Table 7.2. A decision matrix with values and probabilities.

	Positive staff ($p_2 = 65\%$)	Negative staff ($p_1 = 35\%$)
Change platforms	Large gains ($v_2 = 1.0$)	Large losses ($v_1 = 0.0$)
Keep platform	Very small losses ($v_2 = 0.7$)	Small losses ($v_1 = 0.3$)

The information on probabilities is often weighted into an expression that is called the expected value and which is calculated by weighting the values with the probabilities similarly to the multi-criteria analyses in Chapters 5 and 6. In this case, using E to denote the expected value means that we calculate:

$$E(\text{Strategy}) = p_1 \cdot v_1 + p_2 \cdot v_2$$

for each strategy. If we insert our numbers in the expression above, we will obtain the expected values for the strategies. We then choose the strategy with the highest expected value, which according to the calculations and Figure 7.1 is the strategy to change platforms.

$$E(\text{Change platforms}) = 0.65 \cdot 1 + 0.35 \cdot 0 = 0.65$$

$$E(\text{Keep platform}) = 0.65 \cdot 0.7 + 0.35 \cdot 0.3 = 0.56$$

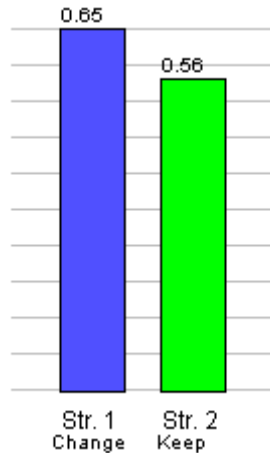


Figure 7.1. Evaluation of decision matrix.

The strategy with the highest expected value is usually perceived as the best. The expected value rule has the great advantage of being clear and easily calculable as long as the probabilities and values are known. The theory of maximising the expected value (or utility) has therefore dominated decision analysis, for instance in financial decision-making when making decisions under risk. It constitutes a so-called normative model of rational choices and is also widespread as a descriptive model of economic behaviour.

7.2. Tree Models

Matrix models are to some extent limiting, and in many situations a tree representation is more transparent because it can represent decisions at several levels, so consequential events and subsequent decisions can also be included. One decision problem often leads to another, and so on; that is, we have decision situations where an event can lead to new events. This is of great importance in, for example, risk analyses.

A decision tree is a structural model that includes the various important components in a decision situation, i.e., the criteria, the strategies, the events, and the consequences, as well as different estimates of the weights of the criteria, the different scenario probabilities, and the values of different consequences. A decision tree thus describes the decision situation more schematically and consists of three types of nodes. In decision nodes (squares), the decision-maker chooses one of several possible strategies. There are then event nodes (circles) where the uncertain outcomes occur, and finally outcome nodes (triangles), which show the outcomes of different paths through the tree. Figure 7.2 shows the same platform example in a tree representation.

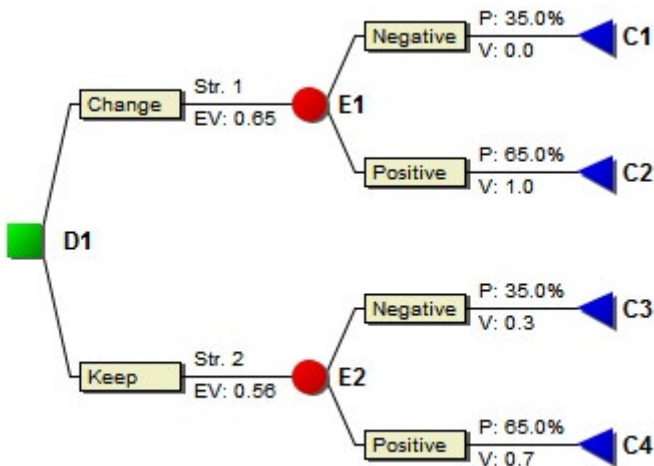


Figure 7.2. Decision tree for decisions under uncertainty.

The strategies are the paths out to the right from the green decision node. In the tree, we assign probability distributions to the event nodes which express the uncertainties of the different possible consequences

of the strategies. These are often estimated using numbers to represent the respective probabilities of the events leading to specific outcomes or sequential events. In Figure 7.2, we have two strategies to choose from. Each will lead to different events with varying consequences. The events will occur with differing probabilities, as shown in the tree. The numbers connected to the outcome nodes represent the values of the different consequences.

A decision tree can express more than a matrix is able to. Suppose, in the platform-changing case, that we have sequential consequences, for example that we get consequential effects of reduced efficiency as the profitability decreases. We can also imagine that the probabilities for different staff reactions will be different, since they are likely dependent on the choice. Figure 7.3 shows the model in an expanded decision tree. Note that some of the probabilities have been added and others changed due to new information as it comes in.

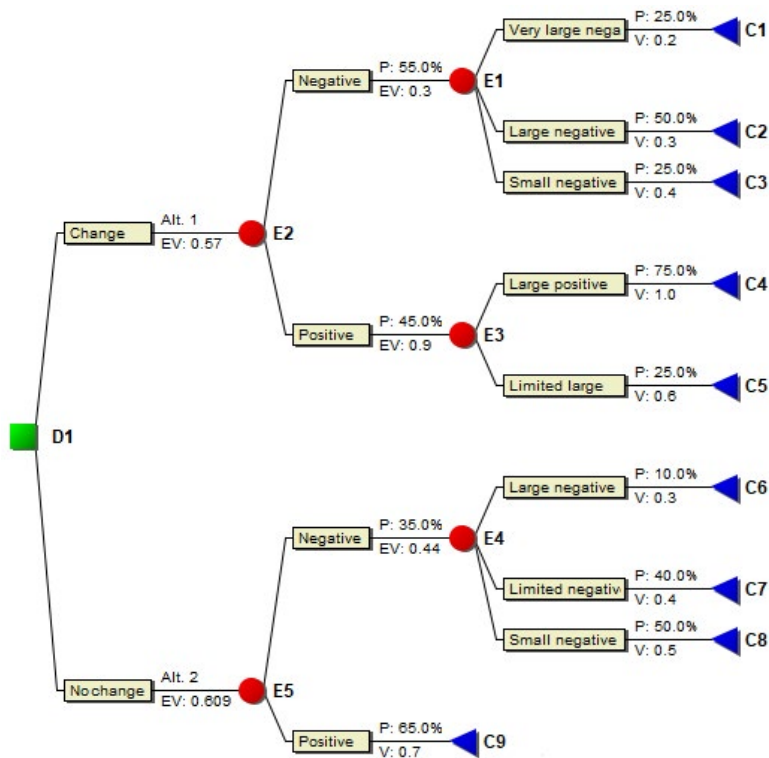


Figure 7.3. Expanded decision tree.

The calculations are similar, but we must consider the partial tree branches:

$$E(\text{Change platforms}) = 0.55 \cdot (0.25 \cdot 0.2 + 0.50 \cdot 0.3 + 0.25 \cdot 0.4) + 0.45 \cdot (0.75 \cdot 1.0 + 0.25 \cdot 0.6) = 0.57$$

$$E(\text{Keep platform}) = 0.65 \cdot 0.7 + 0.35 \cdot (0.10 \cdot 0.3 + 0.40 \cdot 0.4 + 0.50 \cdot 0.5) = 0.609$$

The results can be seen in Figure 7.4.

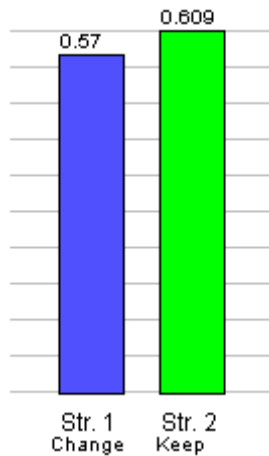


Figure 7.4. Evaluation of two-level decision tree.

Under these conditions, the preferred choice is reversed so that the *keep platform* strategy now has the highest expected value, although only by a slim margin. Through the depiction of the decision tree, it becomes rather straightforward to follow the arcs from a calculated expected value to the underlying probabilities and outcome values, and in the process to derive a sense of the bases of the calculation and the robustness of the differences between expected values.

7.3. Realism in Decision Models

The above model can be useful as a structural component, but keep in mind that it is highly simplified. Always trying to maximise the expected value seems to be a reasonable rule, provided we have the necessary

information. It is also important to understand our preferences and how these relate to the risks involved. We usually talk about risk neutrality when a decision-maker believes that, for example, twice as much of a certain amount of money is perceived as twice as desirable. However, this rule is questionable, because such a linear approach to preferences does not always exist—for organisations or individuals. There are often other factors to consider as well. For example, we might be unwilling to take large risks if we could lose a lot of money, even if the expected value is high.

Many decisions are so important that an undesirable outcome cannot be tolerated, even if it has a low probability. It then seems reasonable to require that if the probability of a very poor outcome is too high, the strategy that could lead to it should be disregarded, even if it exhibits a good expected value. We can say that this is a kind of modified maximin principle. Such safety considerations may be of interest to, for example, insurance companies, who do not want to enter into agreements where the profit opportunities are good but there are non-negligible risks that the outcomes may be so disastrous that the entire company is jeopardised. The expected value maximisation does not take this into account and should be modified on that point.

Therefore, we usually speak of a more general concept of utility, and our own methods are expected to handle utility values that are more complex than simpler financial values. Furthermore, it is already difficult to determine the usefulness of different strategies at the individual level, but when it comes to decisions at the group level, it is even more complex.

Another issue is that just as with criteria weights and strategy values in multi-criteria analyses, a difficulty in risk management with uncertainties is that a decision-maker can seldom give precise numbers, as the above calculations require. Again, the decision-maker usually lacks access to such accurate information and (let us hope) wants to say what he or she knows and nothing else. He or she might want to be able to express claims such as that the value of a certain consequence is between 0.3 and 0.5, that the financial value of another consequence is between EUR 250,000 and EUR 400,000, that the value of a consequence is greater than the value of another consequence, that an event is more probable than another, and so on.

We emphasise once again that in real-world decision-making, we rarely have access to precise information on weightings, probabilities, and values. We most often have to rely on probabilities, which are estimated, often on rather unclear grounds. Sometimes we have frequency probabilities based on statistical data of a reasonable size, but these usually come with uncertainty in the form of confidence intervals. Furthermore, our conception obviously lacks adequate precision, making it impossible to specify fixed probabilities or values. Methods based on fixed numbers thus become difficult to use meaningfully.

Different types of uncertainties basically always appear in real-life situations. Over the years, various methods have been proposed for extending decision analyses to account for this complication. They are usually more mathematically demanding and can be quite cumbersome without a good knowledge of mathematics. Sets of probability distributions, so-called upper and lower probabilities, and interval probabilities are examples of methods for expressing uncertainty in the background data. Within such frameworks we can, for instance, state that a probability lies within a confidence interval, instead of specifying it as an exact number. But even then, there are often large difficulties in assessing information. So, despite the huge assets that are sometimes involved, surprisingly little has been done to carry out proper risk and decision analyses in businesses. However, it is possible to handle impreciseness, although the computational effort required for evaluations increases dramatically.

There are thus, on the one hand, methods that completely lack probability estimates and, on the other hand, methods with over-specified probability estimates. There is consequently a large space (and a great need) for methods that at least allow probability intervals and comparisons to avoid forcing values that are erroneously over-specified. You should specify what you know, and nothing else. Maybe surprisingly, there is often still enough information available to gain a good overview of the situation.

We work with expressions such as the probability of an event being between 10% and 35%, the probability of an event being less than 5%, or one event being more likely than another. Such statements can easily be represented by equations that form a so-called *probability base*. Correspondingly, for example, it can be stated that one consequence is more desirable than another or that the value (or utility) of a consequence

is between 0.3 and 0.7. The value statements here form a so-called *value base* (or utility base).

Since both the probabilities and the utilities are now variables instead of precise values, the expected value calculations give rise to more complex expressions, so-called multilinear expressions. Figure 7.5 shows what this can look like. In the tree, the probabilities are given as intervals instead of precise numbers.

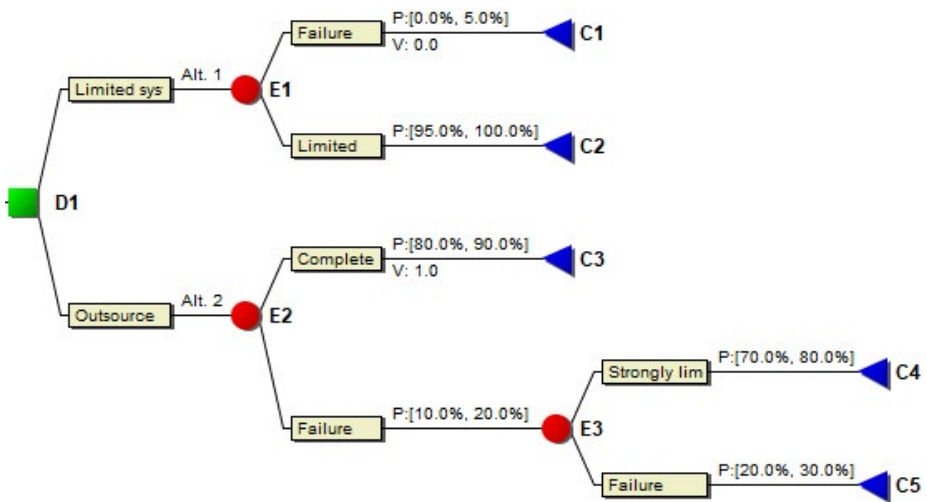


Figure 7.5. A decision tree with intervals.

Now, as before, the expected value is $p_1 \cdot v_1 + \dots + p_n \cdot v_n$ although it is no longer a number but a set of numbers, depending on how the variables (p_i and v_i) are assigned numbers. We will briefly explain how this works. It is not at all necessary to understand this in detail, because one should use software to handle these types of calculations. The details here are unnecessary in the same way that an understanding of how a refrigerator works is unnecessary to be able to use it effectively. If you are not interested in these details, then go ahead and skip to the next section now.

If you want to continue reading here, then first, we will look at some domination concepts defining when one strategy is better than another. We denote the expected utility of strategy X as $E(X)$. By “base”, we mean all the information available in the decision model.

- Strategy A is at least as good as B in the current probability and value bases of $E(A) \geq E(B)$ for all feasible values of the included probabilities and utility values.
- Strategy A is better than B in the current probability and value bases if it is at least as good as B and $E(A) > E(B)$ for any allowable value of the included probabilities and utility values.

If there is only one strategy that is the best one, then that should be selected, but normally, we have overlaps between strategies that make it more difficult to directly determine which strategy is the best. In this case, it is appropriate to consider the number of values (the so-called support) a strategy has by examining for how many different assignments of variables in the bases (which contain all the possible numbers that represent the perceptions we have) the different strategies come out as the best. If, for example, strategy A is better than B for 2% of the feasible numbers, and B is better than A for 98%, it seems reasonable to choose B over A because it is much more likely that B is in fact the best strategy; that is, in a much larger proportion of all possible cases, B is a better choice. To facilitate these calculations, you need quite an elaborate computer program.

We will now illustrate a more difficult decision situation with a simple example (which also demonstrates that small problems are already difficult to solve without structured methods).

For instance, assume that we are about to implement a new IT system, and that we have two main strategies to choose from. One strategy consists of building up from scratch a somewhat limited system with internal resources, and the other consists of outsourcing a larger project for an extended system. After an analysis, we find that the risk of a total failure in internal development is less than 5%. We then have at least a 95% probability of obtaining a working system, albeit a somewhat limited one. If we outsource, the probability of subsequently being forced to carry out an internal development project and obtaining an even more limited system is 10–20%. This means that the risk of total failure is 20–30%.

As you can see, the problem is structurally relatively simple, but it is still difficult to give a reasonable and well-justified recommendation by just considering this *prima facie*. The processing problems we are facing

are that: (i) we lack access to accurate probabilities, as the estimates in the data are necessarily uncertain, and (ii) a reasonable value scale is not easy to define.

If we instead ignore the requirement for unrealistic precision, we can rank the consequences: i) A full-scale system is better than an initially limited system, ii) An initially limited system is better than a system that has even more limited functionality and that must be built quickly, and iii) A system with even more limited functionality is better than a failed project.

This problem is short, but nevertheless it is not clear which strategy is the best simply from reading it. To be able to analyse it more structurally, we must start modelling the problem as a decision tree. In Figure 7.6 (which is the same as Figure 7.5), we can see the two strategies and their respective consequences in such a tree. The probabilities are stated at the edges of the tree as the intervals we specified above. Here we also have an underlying value scale for the consequences between 0 and 1. We give a value of 0 to the worst consequence (complete failure) and a value of 1 to the best (full functionality).

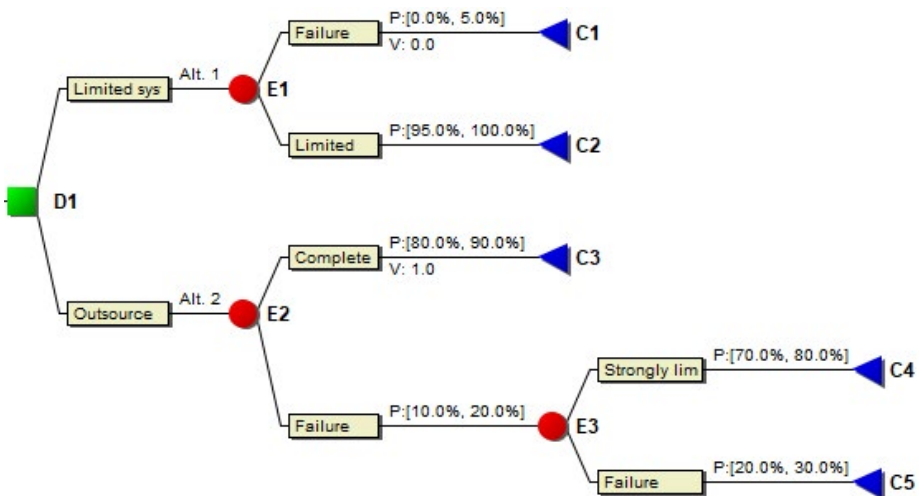


Figure 7.6. The decision tree modelling of the example.

After modelling the problem, an evaluation can be made. We use the expected value in the slightly modified version that we described above. Since the problem contains both intervals and comparisons, it is again

slightly complicated to calculate manually, particularly as there are now dependencies between the two strategies. We therefore use the DecideIT tool for the analysis, which also gives us some additional information. The result of this analysis can be seen in Figure 7.7.

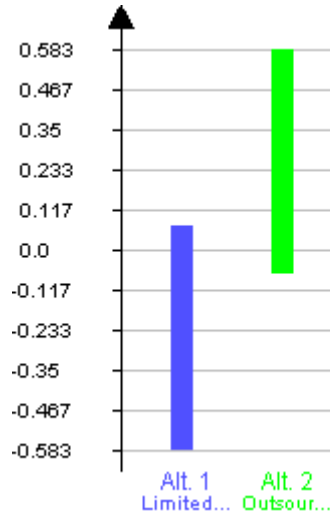


Figure 7.7. An initial evaluation of the decision situation.

What we see in the figure is how well the strategies perform relative to each other given the values in the bases that we have defined with our interval and relationship expressions. The higher the bar, the better the strategy it represents. The leftmost bar represents the strategy of building the system internally, and the rightmost one represents the strategy of outsourcing. It is now easy to see that the outsourcing strategy is clearly better than that of building the system internally. There is a small overlap, but it has no practical significance since it represents only a small part of the information bases and therefore can safely be ignored. It is thus evident that the best strategy is outsourcing, based on the information that has been provided. The situation is furthermore very stable in relation to changes in the input information. Decision problems like this, and many considerably more complicated ones, are easily manageable if we perform the analysis in this way. What was initially difficult to see becomes clear through this method, without the addition of any information that we do not actually have.

7.4. Sensitivity Analyses

A sensitivity analysis is indispensable in a robust decision-making process. The decision process involves continuously questioning the model. Have we really modelled and analysed the problem reasonably correctly? We should ask ourselves whether we have inferred the most relevant strategies, consequences, scenarios, and criteria. But the most common way to check the result is to vary the input probabilities and utilities as well as (if applicable) the criteria weights, to see how sensitive the decision is to changes. This is called sensitivity analysis and is normally used to find out so-called critical values, i.e., the limits within which weights, probabilities, and values must remain in order for the result not to change.

This is achieved by letting several variables alternate between the most and least likely, but still valid, numbers and studying how this affects the decision. Starting with the simplest model with fixed numbers for probabilities and values, we first look at the values of the consequences. The value of each consequence and the probability of each event must therefore be systematically varied upwards and downwards. If you have structural information, such as the knowledge that one consequence is worth more than another, there are dependencies that cause the variables to co-vary, which must also be considered.

Reconsider the platform change example. The sensitivity analysis could start with, for example, the probabilities being varied to see how this would change the decision. For instance, we might want to change the value of smaller losses to 0.2 and 0.4, respectively, (from 0.3) to see if this has any effect.

Table 7.3. A modified decision matrix with values and probabilities.

	Positive staff ($p_2 = 65\%$)	Negative staff ($p_1 = 35\%$)
Change platform	Big gains ($v_2 = 1.0$)	Big losses ($v_1 = 0.0$)
Keep platform	Very small losses ($v_2 = 0.7$)	Small losses ($v_1 = 0.2$ and 0.4 respectively)

If we insert our new numbers into the expression above, we will obtain new values for the second expected value.

$$E(\text{Change platform}) = 0.65 \cdot 1 + 0.35 \cdot 0 = 0.65, \text{ as before}$$

$$E(\text{Keep platform}) = 0.65 \cdot 0.7 + 0.35 \cdot 0.2 = 0.52 \text{ and } 0.65 \cdot 0.7 + 0.35 \cdot 0.4 = 0.60$$

It is still best to change the platform according to the calculation, so in this sense, the decision is quite stable if we consider this variable. One problem, however, is that it is only realistic to perform the sensitivity analysis in this way when we have few variables. As soon as we add more variables that we want to co-vary, it quickly becomes unmanageable. We can handle more complicated sensitivity analyses much more easily if, for example, we already have interval statements and comparisons for probabilities, values, and weights from the start. In that case, we just allow them to assume all possible values given the information we have, by means of a software tool.

Let us return to the example of outsourcing shown again in Figure 7.8 (which repeats Figure 7.5).

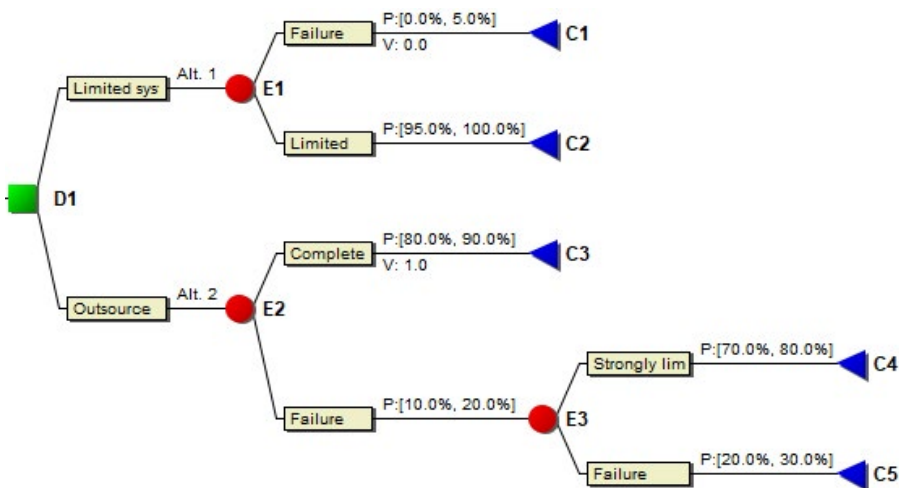


Figure 7.8. The decision tree shown again.

Earlier, we saw that the strategy of outsourcing was the best one. Now we will analyse the sensitivity of the information bases. We can do this in several ways. A tornado diagram is one method of showing how a

strategy’s expected value varies when the parameters involved are varied. The diagram shows, for example, how variations in the probabilities in a decision problem affect its end result. The diagram is sorted so that the probabilities with the greatest effect are at the top, with the remaining probabilities below them in decreasing order of influence. The width of each parameter indicates the influence on the expected value. The diagram thus resembles a tornado, as shown in Figure 7.9. Similarly, we can see how changes in the values and weights affect the result.

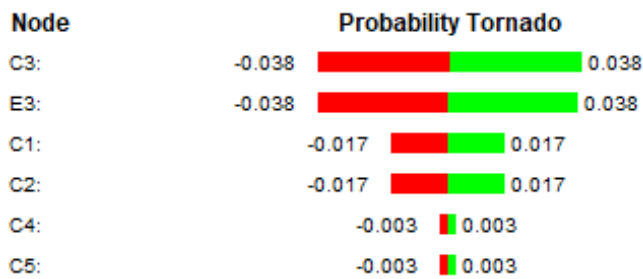


Figure 7.9. A tornado diagram.

An even better way to study the stability of decisions is to look at where the most important numbers reside. For example, when you specify an interval, it is likely that not all numbers in the interval are of equal importance, but rather that the numbers towards the middle of the interval add more “decision weight” to the problem. This could be seen as a kind of function representing the strengths that we assign to the different values. Figure 7.10 shows an example of how we might believe in the probability of success in building a limited system.

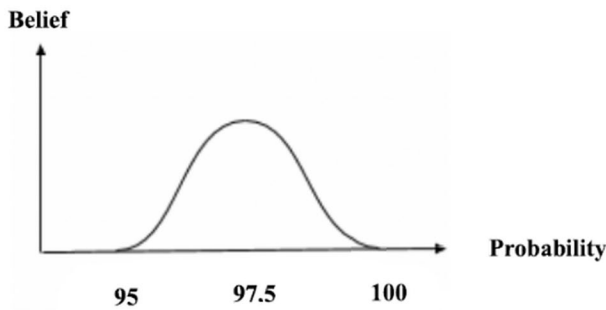


Figure 7.10. How important values can be distributed.

What we would like to know is how the decision situation looks when you remove the numbers closer to the interval borders and instead consider only the central ones. This is called contraction and can be done systematically with all variables, right up to the single number where you have the strongest beliefs in the values. In Figure 7.10, the strongest belief is at a probability of 97.5%. Figures 7.11 and 7.12 show what happens to the evaluation of the strategies when we systematically and continuously peel away the border numbers from left to right (increasing degree of contraction).

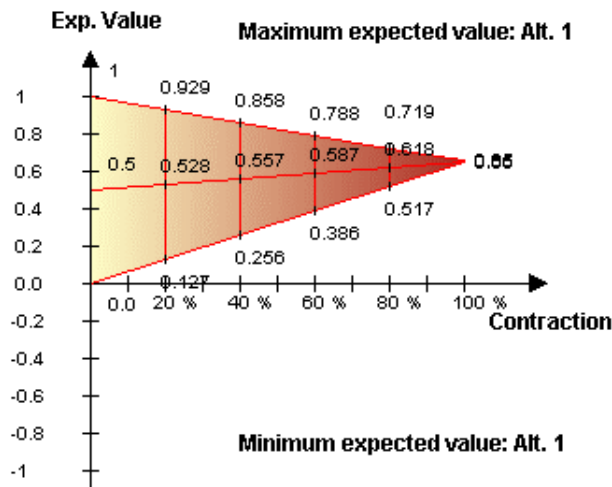


Figure 7.11. Focusing on central interval numbers.

For the first strategy (building a limited system), you can see how the result of the evaluation from the beginning (at 0% contraction) can be between 0 and 1. At 40% contraction, it is between 0.256 and 0.858, and at full contraction, it is 0.65. Correspondingly (but not shown), the values for the second strategy (outsourcing) are initially between 0.8 and 0.98, and finally reach 0.8875 at 100% contraction. In this way, we see that the decision is very stable, and that the second strategy is significantly better than the first. We can also directly study the differences between the strategies, as shown in Figure 7.12.

In Figure 7.12, we can see how the second strategy compares to the first. The greater the area of the triangle that lies above the x-axis, the better the second strategy is. We can see that that difference is quite dramatic. This means that the decision to choose the second strategy is

very stable. In some cases, it may be more difficult to determine which strategy is the best (for instance, when the cone area in the figure is centred close to the x-axis). In that case, additional information must be gathered in order to determine the most suitable course of action. Through the tornado diagrams discussed above, we can then see which information has the greatest impact, so that we can obtain a better picture of how to allocate resources for continued analysis.

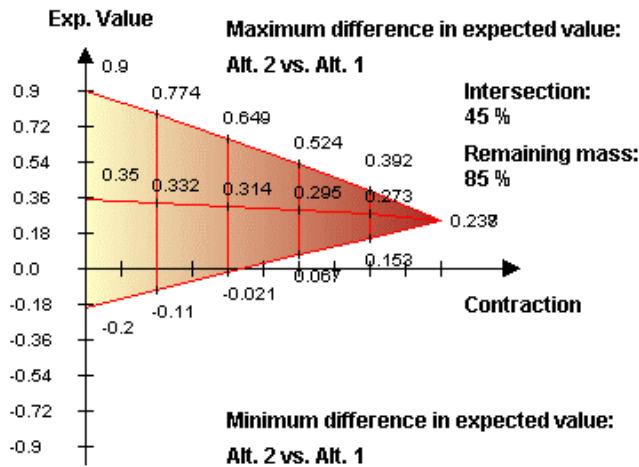


Figure 7.12. Comparison of the strategies.

An important observation can be made here from what is called the remaining mass in the figures. This expresses more exactly how strong the difference between the alternatives is. In this case, it is 85%, which means that 85% of the information points to the second strategy being the best and only 15% points to the first, i.e. the result is very stable. We will not go through all of the details here, but we will refer to more information that can be found in the tips for further reading and the appendix. We hope that this has shown the advantages of performing sensitivity analyses, and that with the right tools they are both manageable and informative. Finally, we will consider tools.

7.5. Tool Support

Advanced decision tools are easily able to handle all of the components we have discussed in this chapter, such as criteria and sub-criteria,

weightings, strategies, events and probabilities, and values (utilities). In this way, we obtain a basis for the decision, where we can combine decision trees and criteria trees for one and the same decision problem. There are different types of tools that we can use to support this process, but we should use those that do not limit what we want to accomplish. Where information uncertainty is considered, we must be able to handle quantitative information in a reasonable manner. We must also be able to handle soft information such as rankings that can be difficult or even impossible to express in numbers. Furthermore, we need to be able to mix hard and soft information and to view the decision problem from different perspectives. The tools should be able to support all of this, point out the weak and strong strategies, and explain their characteristics.

The program we have used in this book is DecideIT, which is available at a discounted rate for universities and other public entities and at a standard rate for all other users. A one-year licence with full functionality is supplied for free with this book. Refer to the appendix for instructions and licence information. See also the paragraph below on further reading for more information.

7.6. Chapter Summary

We have reviewed various aspects of decision analysis and how it can be used to achieve rational, transparent, and stable decision situations. We have in this chapter mainly dealt with probabilities, and discussed how they influence decision outcomes. An important observation is that, as in multi-criteria analyses, it is practically untenable to use precise numbers for what is in reality uncertain information and where in practice we have neither full insight on the future, nor on our actual preferences.

Therefore, we must use models that are useful in practice, that allow us to provide imprecise information but still manage it, and that give us sensible directives. Otherwise, our decision-making will only be based on diffuse gut feelings and impulses. We have gone through various aspects of probabilities and risks, and how these can be systematically handled. We have also considered instances when even rather vague information can satisfactorily give an insight into which decision is the best.

In the following chapters, we will take a closer look at how to systematically incorporate decision-making into a project management process.

7.7. Reading Tips

These tips pertain to Chapters 5–7.

Transcending Business Intelligence addresses different aspects of decision and risk analysis in a simple manner, and also goes through some examples of how the analyses can be used in practice.

- Borking, Kjell; Danielson, Mats; Davies, Guy; Ekenberg, Love; Idefeldt, Jim and Larsson, Aron (2022). *Transcending Business Intelligence* (3rd edn). Stockholm: Sine Metu. Available free of charge from www.preference.nu.

A more comprehensive but more difficult book on the subject is *Deliberation, Representation, Equity*, which carefully examines uncertainty in decisions. It also discusses how this can work when people participate in political decision-making.

- Ekenberg, Love; Hansson, Karin; Danielson, Mats; Cars, Göran et al. (2017). *Deliberation, Representation, Equity: Research Approaches, Tools and Algorithms for Participatory Processes*. Cambridge: Open Book Publishers. <https://doi.org/10.11647/OBP.0108>.

Integrated Catastrophe Risk Modelling deals with various aspects of risks. The book focuses on flood risks but is applicable more generally. It introduces several useful methods for risk management.

- Amendola, Aniello; Ermolieva, Tatiana; Linnerooth-Bayer, Joanne and Mechler, Reinhard (eds) (2013). *Integrated Catastrophe Risk Modelling: Supporting Policy Processes*. Cham: Springer. <https://doi.org/10.1007/978-94-007-2226-2>.

A good overview of decision analysis can be found in the article “Decision Analysis: An Overview”. It is a bit old but is written by one of the big names in the area and is still very relevant.

- Keeney, Ralph (1982). Decision Analysis: An Overview. *Operations Research* 30 (5), pp. 803–838, <https://doi.org/10.1287/opre.30.5.803>.

Kahneman's book is an excellent and systematic description of people's shortcomings regarding rational thinking and reasoning. In 2002, Kahneman received the Nobel Prize in Economics (to be precise, the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel).

- Kahneman, Daniel (2012). *Thinking, Fast and Slow*. London: Penguin Books. <https://doi.org/10.1111/j.1539-6975.2012.01494.x>.

If you want to learn more about the tool DecideIT, you can read the following two papers by Danielson et al. They go through the fundamentals of the tool without delving too deeply into mathematics.

- Danielson, Mats; Ekenberg, Love and Larsson, Aron (2019). DecideIT 3.0: Software for Second-Order Based Decision Evaluations. Proceedings of the Eleventh International Symposium on Imprecise Probabilities. *Proceedings of Machine Learning Research* 103, pp. 121–124.
- Danielson, Mats; Ekenberg, Love and Larsson, Aron (2020). A second-order-based decision tool for evaluating decisions under conditions of severe uncertainty. *Knowledge-Based Systems* 191, 105219. <https://doi.org/10.1016/j.knosys.2019.105219>.

If you really want to understand the details of the program and its mathematical background, you can refer to Danielson and Ekenberg's patent description.

- Danielson, Mats and Ekenberg, Love (2004). A Method for Decision and Risk Analysis in Probabilistic and Multiple Criteria Situations under Incomplete Information, U.S. patent 7257566.

Almeida et al.'s book is an overview of the multi-criteria area that goes through various processes and techniques.

- de Almeida, Adiel; Ekenberg, Love; Scarf, Philip; Zio, Enrico and Zuo, Ming J. (2022) *Multicriteria and Optimization Models for Risk, Reliability, and Maintenance Decision Analysis*. Cham: Springer. <https://doi.org/10.1007/978-3-030-89647-8>.

A somewhat older, but still very good, overview of the multi-criteria area has been written by Belton and Stewart.

- Belton, Valerie and Stewart, Theo (2002). *Multiple Criteria Decision Analysis: An Integrated Approach*. Cham: Springer. <https://doi.org/10.1007/978-1-4615-1495-4>.

Hubbard provides a good critical review of the area of risk management and discusses lots of interesting examples.

- Hubbard, Douglas (2009). *The Failure of Risk Management: Why It's Broken and How to Fix It*. Hoboken, NJ: John Wiley & Sons. <https://doi.org/10.1002/9781119198536>.

The Feeling of Risk is an entertaining overview of how we perceive risks by one of the most knowledgeable people in the field.

- Slovic, Paul (2010). *The Feeling of Risk: New Perspectives on Risk Perception*. London: Earthscan. <https://doi.org/10.4324/9781849776677>.

At the website www.sipta.org, there is lots of information on different ways to represent uncertainty in probabilities. Typically, the material is quite difficult to read but the website gives a good overview of the subject area.

The program we have used and reference in this book is DecideIT, which is complimentary with every copy of the book and can be downloaded from www.preference.nu/digitrans. The appendix describes how to install and use the program for structured decision-making. You can read more about the program at www.preference.nu. More about decision-making programs in general can be found at, for example, Wikipedia. Search for “decision-making software” or “decision-analytic software” for an up-to-date overview.

